

ALI RAHIMPOUR JOUNGHANI, PH.D.

Research Scientist, Stanford University School of Medicine

Co-Founder and Chief Scientific Officer, Walnut Inc.

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RESEARCH INTERESTS

Development of wearable functional near-infrared spectroscopy (fNIRS) platforms and computational methods for precision mental health. My work combines multimodal neuroimaging (fNIRS, EEG, fMRI) with machine learning to characterize individual brain function, to guide neurofeedback-based cognitive interventions for children with ADHD, and to study clinical populations including PTSD, Prader-Willi syndrome, and Alzheimer's disease. Current work builds self-supervised foundation models for optical neuroimaging to enable scalable brain-state discovery across tasks, populations, and real-world contexts.

Keywords: cognitive and computational neuroscience · wearable neurotechnology · precision psychiatry · advanced neuroimaging · machine learning and foundation models · biomedical engineering

ACADEMIC AND PROFESSIONAL APPOINTMENTS

Research Scientist (part-time) 2026 – present

Department of Psychiatry and Behavioral Sciences, Stanford University School of Medicine

– C-Brain Lab (PI: S. M. Hadi Hosseini, Ph.D.)

Co-Founder and Chief Scientific Officer 2026 – present

Walnut Inc.

– Principal Investigator, NIH STTR Phase I (R41) award to develop a wearable fNIRS platform for precision psychiatry

Postdoctoral Scholar 2022 – 2025

Department of Psychiatry and Behavioral Sciences, Stanford University School of Medicine

– NIH T32 Postdoctoral Fellow, Center for Interdisciplinary Brain Sciences Research (2024 – 2025)
– C-Brain Lab; Mentor: S. M. Hadi Hosseini, Ph.D.

EDUCATION

Ph.D., Psychological Sciences 2017 – 2022

University of California, Merced, CA

– Dissertation: *The effect of contextually specific, action-based timing behavior on brain responses*
– Committee: Heather Bortfeld (Chair), Rose Scott, Ramesh Balasubramaniam, Elif Isbell

M.S., Quantitative Methods (Quantitative Psychology) 2021

University of California, Merced, CA

M.S., Biomedical Engineering 2013 – 2016

University of Tehran, Tehran, Iran

– Thesis: *Investigation of fNIRS application for infant cerebral hemodynamic monitoring: A report of data analysis for feature extraction and infant classification into healthy and unhealthy*
– Advisors: Hossein Ahmadi Noubari and Seyed Kamaledin Setarehdan

GRANTS AND FELLOWSHIPS

2025	Principal Investigator, NIH Small Business Technology Transfer (STTR) Phase I Award (R41), Walnot Inc. — wearable fNIRS platform for precision psychiatry in children with ADHD
2024 – 2025	NIH T32 Postdoctoral Fellowship, Center for Interdisciplinary Brain Sciences Research, Stanford University School of Medicine
2021	Hatano Cognitive Development Research Fellowship, University of California, Merced
2020	Developmental Student Research Award, University of California, Merced
2018 – 2022	Bobcat Research Support Fellowship (summers), University of California, Merced
2019	Graduate Student Association Travel Grant, University of California, Merced
2017	Research Grant, Mofid Children's Hospital, Tehran, Iran
2016	Research Grant, Cognitive Sciences and Technologies Council, Iran

HONORS AND AWARDS

2026	Top Innovator Honoree, Venture Summit West 2026 (youngStartup Ventures) — Walnot Inc.
2026	Selected Startup, Ferment 2026, Stanford Biotech Group — Walnot Inc.
2025	Outstanding Oral Presentation Award, UK fNIRS Conference, Cambridge, UK — “Neuroscience of mathematics learning in children: Application of fNIRS in South Africa”
2025	Finalist Pitch Presenter, Stanford Founders Demo Day
2024	Finalist, Wearable Technologies Innovation World Cup
2024	Finalist, Stanford High Impact Technology (HIT) Fund

PUBLICATIONS

*Bold denotes the author (also published as **Rahimpour, A.** and **Jounghani, A. R.**). Full list on Google Scholar.*

Peer-Reviewed Journal Articles

1. Aitken, A., **Jounghani, A. R.**, Carbonell, L. M., Tadeo, D., Kumar, A., Crawford, S., ... & Hosseini, S. H. (2025). Naturalistic fNIRS assessment reveals decline in executive function and altered prefrontal activation following social media use in college students. *Scientific Reports*, 15(1), 36960.
2. Fonseca, K., Nemati, P., **Jounghani, A. R.**, Henning, E., & Soltanlou, M. (2025). Neural correlates of fraction magnitude processing in high and low achieving primary school children in South Africa. *Cognitive Development*, 76, 101636.
3. Yücel, M. A., Luke, R., Mesquita, R. C., von Lühmann, A., Mehler, D. M., Lühns, M., ... & Zemanek, V. (2025). fNIRS reproducibility varies with data quality, analysis pipelines, and researcher experience. *Communications Biology*, 8(1), 1149.

4. O'Connell, S. R., **Rahimpour Jounghani, A.**, Papadopoulos, J. M., Bortfeld, H., & Goldsworthy, R. L. (2025). Investigating hemodynamic patterns during beat processing in cochlear implant users: Insights from a finger tapping study. *Auditory Perception & Cognition*, 1–25.
5. **Rahimpour Jounghani, A.**, Kumar, A., Moreno Carbonell, L., Aguilar, E. P. L., Picardi, T. B., Crawford, S., ... & Hosseini, S. M. (2025). Wearable fNIRS platform for dense sampling and precision functional neuroimaging. *npj Digital Medicine*, 8, 271. <https://doi.org/10.1038/s41746-025-01690-3>
6. Gozdas, E., Kim, T., **Jounghani, A. R.**, & Hosseini, S. H. (2025). Protocol to predict time perception bias from cortical neurite microstructures in healthy older adults. *STAR Protocols*, 6(2), 103756.
7. Stute, K., Gossé, L. K., Montero-Hernandez, S., Perkins, G. A., Yücel, M. A., Cutini, S., ... & Zainodini, H. (2025). The fNIRS glossary project: A consensus-based resource for functional near-infrared spectroscopy terminology. *Neurophotonics*, 12(2), 027801.
8. **Rahimpour Jounghani, A.**, Gozdas, E., Dacorro, L., Avelar-Pereira, B., Reitmaier, S., Fingerhut, H., Hong, D. S., Elliott, G., Hardan, A. Y., Hinshaw, S. P., & Hosseini, S. M. H. (2024). Neuromonitoring-guided working memory intervention in children with ADHD. *iScience*, 27(11), 111087. <https://doi.org/10.1016/j.isci.2024.111087>
9. Kim, T., **Jounghani, A. R.**, Gozdas, E., & Hosseini, S. H. (2024). Cortical neurite microstructural correlates of time perception in healthy older adults. *Heliyon*.
10. **Jounghani, A. R.**, Backer, K. C., Vahid, A., Comstock, D. C., Zamani, J., Hosseini, H., ... & Bortfeld, H. (2024). Investigating the role of auditory cues in modulating motor timing: Insights from EEG and deep learning. *Cerebral Cortex*, 34(10), bhac427.
11. **Rahimpour Jounghani, A.**, Lanka, P., Pollonini, L., Proksch, S., Balasubramaniam, R., & Bortfeld, H. (2023). Multiple levels of contextual influence on action-based timing behavior and cortical activation. *Scientific Reports*, 13(1), 7154.
12. Ghafari, T., **Jounghani, A. R.**, & Esteky, H. (2023). Where and when matter in visual recognition. *Attention, Perception, & Psychophysics*, 85, 404–417. <https://doi.org/10.3758/s13414-022-02607-y>
13. **Rahimpour, A.**, Pollonini, L., Comstock, D., Balasubramaniam, R., & Bortfeld, H. (2020). Tracking differential activation of primary and supplementary motor cortex across timing tasks: An fNIRS validation study. *Journal of Neuroscience Methods*, 108790.
14. Ebrahimzadeh, E., Shams, M., **Jounghani, A. R.**, Fayaz, F., Mirbagheri, M., Hakimi, N., ... & Soltanian-Zadeh, H. (2020). Localizing confined epileptic foci in patients with an unclear focus or presumed multifocality using a component-based EEG-fMRI method. *Cognitive Neurodynamics*, 1–16.
15. **Rahimpour, A.**, Noubari, H. A., & Kazemian, M. (2018). A case-study of NIRS application for infant cerebral hemodynamic monitoring: A report of data analysis for feature extraction and infant classification into healthy and unhealthy. *Informatics in Medicine Unlocked*, 11, 44–50.

Manuscripts Under Review

1. **Rahimpour Jounghani, A.**, & Hosseini, S. M. H. (submitted). What a brain measure must show before it can guide psychiatric treatment. *npj Digital Medicine* (Perspective).

2. Faghihzadeh, S.†, **Rahimpour Jounghani, A.†**, Moreno Carbonell, L., Mei, M., Izuzkiza, C., & Hosseini, S. M. H. (submitted). Wearable fNIRS for at-home deep phenotyping with autocorrelation-aware inference: ADHD application. *BME Frontiers*. †Equal contribution.
3. Balters, S., Lefevre, S., **Rahimpour Jounghani, A.**, Kaysen, D., & Reiss, A. L. (submitted). Toward a triple network connectivity signature of PTSD using resting-state fNIRS. *Clinical Psychological Science*.
4. Balters, S., Lefevre, S., Brier, Z. F. M., **Rahimpour Jounghani, A.**, & Kaysen, D. (submitted). Threat-selective prefrontal responses to social cues in PTSD and their association with perceived isolation and hyperarousal. *Progress in Neuro-Psychopharmacology & Biological Psychiatry*.
5. Aguilar, E. P., **Rahimpour Jounghani, A.**, Hosseini, S. M. H., & Hall, S. S. (submitted). Feasibility of naturalistic fNIRS during skin-picking in Prader–Willi syndrome: Environmental control of behavior and exploratory cortical analyses. *Journal of Psychiatric Research*.

Manuscripts in Preparation

1. Moreno Carbonell, L.†, **Rahimpour Jounghani, A.†**, Faghihzadeh, S., & Hosseini, S. M. H. (in preparation). Inter-device reliability of a wearable wireless fNIRS headband compared with an fMRI-validated NIRx NIRSport2 system. †Co-first authors.
2. **Rahimpour Jounghani, A.**, Vahid, A., Bhati, S., & Hosseini, S. M. H. (in preparation). Why fNIRS still lacks a general-purpose foundation model, and what one would require.
3. **Rahimpour Jounghani, A.**, et al. (in preparation). Precision Wearable: An open MATLAB toolbox for dense-sampling analysis of wearable fNIRS and EEG.
4. **Rahimpour Jounghani, A.**, Hosseini, S. M. H., et al. (in preparation). Near-infrared spectroscopy in perioperative medicine: Resolving the controversy through the lens of functional neuroimaging. Invited review with perioperative medicine collaborators.

Peer-Reviewed Conference Proceedings

1. Bowden, A. K., Kumar, A., Crawford, S., **Jounghani, A. R.**, Carbonell, L. M., & Hosseini, S. H. (2026, April). fNIRS for everyone: Toward accessible, inclusive, and scalable neuroimaging. In *Optics and the Brain* (pp. BS5A-1). Optica Publishing Group.
2. **Rahimpour, A.**, Dadashi, A., Soltanian-Zadeh, H., & Setarehdan, S. K. (2017, April). Classification of fNIRS-based brain hemodynamic response to mental arithmetic tasks. In *2017 3rd International Conference on Pattern Recognition and Image Analysis (IPRIA)* (pp. 113–117). IEEE.
3. Jahani, S., Berivanlou, N. H., **Rahimpour, A.**, & Setarehdan, S. K. (2015, November). Attention level quantification during a modified Stroop color word experiment: An fNIRS based study. In *2015 22nd Iranian Conference on Biomedical Engineering (ICBME)* (pp. 99–103). IEEE.

Preprints

1. Kumar, A., Crawford, S., Le, T. C., **Jounghani, A. R.**, Carbonell, L. M., Capps, A. S., ... & Bowden, A. K. (2024). NIRDuino: A modular, Bluetooth-enabled, Android®-configurable fNIRS system with dual-intensity mode built on Arduino®. *medRxiv*.
<https://doi.org/10.1101/2024.12.20.24318425>

PRESENTATIONS

Invited Talks

1. **Rahimpour, A.** (2023, July). Wearable brain interfaces for health. *eWEAR Seminar Series*, Stanford University, Stanford, CA.

Conference Presentations and Published Abstracts

1. **Rahimpour Jounghani, A.**†, Faghihzadeh, S., Mei, M.†, Moreno Carbonell, L., Izuzkiza, C., & Hosseini, H. (2026, May). At-home naturalistic neuroimaging: A pilot study of wearable fNIRS during daily life activities. *Wu Tsai Neurosciences Institute Retreat*, Stanford University, CA.
†Presenters.
2. Faghihzadeh, S.†, **Rahimpour Jounghani, A.**†, Moreno Carbonell, L., & Hosseini, S. M. H. (2026, May). At-home deep phenotyping with a wearable fNIRS platform: A pilot study in a single ADHD case using dense sampling. *Wu Tsai Neurosciences Institute Retreat*, Stanford University, CA. †Equal contribution.
3. Moreno Carbonell, L., Izuzkiza, C., **Rahimpour Jounghani, A.**, Faghihzadeh, S., Mei, M., & Hosseini, S. M. H. (2026, May). Precision mental health platform for ecological studies. *Wu Tsai Neurosciences Institute Retreat*, Stanford University, CA.
4. Soltanlou, M., Fonseca, K., Bezuidenhout, H. S., Nemati, P., **Rahimpour Jounghani, A.**, Borj Khani, H., & Henning, E. (2025, September). Neuroscience of mathematics learning in children: Application of fNIRS in South Africa. *UK fNIRS Conference*, Cambridge, UK. [Outstanding Oral Presentation Award]
5. Aguilar, E. P., **Rahimpour Jounghani, A.**, Hosseini, H., & Hall, S. (2025, June). Functional near-infrared spectroscopy of skin picking behavior in Prader-Willi syndrome. *International Prader-Willi Syndrome Conference*, Phoenix, AZ.
6. **Rahimpour Jounghani, A.**, Moreno Carbonell, L., Kumar, A., Bowden, A. K., & Hosseini, H. (2025, May). Advancing individualized functional neuroimaging with a wearable fNIRS platform: A proof-of-concept study on dense-sampling and reliability. *American Psychiatric Association Annual Meeting*, Los Angeles, CA.
7. **Rahimpour Jounghani, A.**, Gozdas, E., Dacorro, L., & Hosseini, S. M. H. (2024). Neuromonitoring-guided working memory intervention in children with ADHD. *Society of Biological Psychiatry Annual Meeting*, Austin, TX.
8. **Rahimpour, A.**, Kumar, A., Crawford, S., Bowden, A., & Hosseini, H. (2023, October). Wearable fNIRS platform for precision mental health. *Innovation and Discovery Expo*, Stanford University, Stanford, CA.
9. **Rahimpour, A.**, Moreno Carbonell, L., Lopez Aguilar, P., Picardi, T., & Hosseini, H. (2023, July). Portable fNIRS platform for cognitive enhancement and precision mental health. *Center for Mind, Brain, Computation and Technology*, Wu Tsai Neurosciences Institute, Stanford, CA.
10. O'Connell, S. R., **Rahimpour, A.**, Papadopoulos, J., Nowlen, F., Bortfeld, H., & Goldsworthy, R. (2022, October). Cochlear implant users and normal-hearing listeners have contrasting hemodynamic changes during auditory beat processing despite similar behavioral responses. *Biennial Meeting of the Society for Functional Near-Infrared Spectroscopy (SfNIRS)*, Boston, MA.

11. Isaiah, A., Matthew, N., **Rahimpour, A.**, Pereira, K., & Bortfeld, H. (2022, October). Tracking prefrontal cortex activation predicts impulsiveness in children with sleep disordered breathing. *Biennial Meeting of the Society for Functional Near-Infrared Spectroscopy (SfNIRS)*, Boston, MA.
12. O'Connell, S. R., **Rahimpour, A.**, Papadopoulos, J., Nowlen, F., Bortfeld, H., & Goldsworthy, R. (2022, February). Tracking motor and auditory cortex activation during synchronous tapping in cochlear implant users and normal-hearing listeners: An fNIRS validation study. *45th Annual Mid Winter Meeting of the Association for Research in Otolaryngology (ARO)*, San Jose, CA.
13. **Rahimpour, A.**, Pollonini, L., Comstock, D., Balasubramaniam, R., & Bortfeld, H. (2021, October). The effect of study design context on timing behavior and cortical brain activation: fNIRS alternating vs. block design study. *Society for Functional Near-Infrared Spectroscopy Virtual Conference*.
14. Ghafari, T., **Rahimpour, A.**, & Esteky, H. (2021, September). Memory performance is heterogeneous around the visual field. *Bernstein Conference on Computational Neuroscience*.
15. **Rahimpour, A.**, Comstock, D., Pollonini, L., Balasubramaniam, R., & Bortfeld, H. (2021, January). Tracking differential activation of primary and supplementary motor cortex across timing tasks: An fNIRS validation study. *4th Annual Boston University Neurophotonics Symposium*, Boston, MA.
16. **Rahimpour, A.**, Comstock, D., Pollonini, L., Balasubramaniam, R., & Bortfeld, H. (2021, January). The effect of study design context on timing behavior and cortical brain activation: fNIRS alternating vs. block design study. *4th Annual Boston University Neurophotonics Symposium*, Boston, MA.
17. **Rahimpour, A.**, Pollonini, L., Comstock, D., Balasubramaniam, R., & Bortfeld, H. (2019, November). Tracking differential activation of primary and supplementary motor cortex across timing tasks: An fNIRS validation study. *60th Annual Meeting of the Psychonomic Society*, Montreal, QC, Canada.
18. Ebrahimzadeh, E., Shams, M., **Rahimpour Jounghani, A.**, Fayaz, F., Mirbagheri, M., Hakimi, N., ... & Soltanian-Zadeh, H. (2019). Epilepsy presurgical evaluation of patients with complex source localization by a novel component-based EEG-fMRI approach. *Iranian Journal of Radiology*, 16(Special Issue). [Published abstract]
19. **Rahimpour, A.**, Pollonini, L., Comstock, D., Balasubramaniam, R., & Bortfeld, H. (2019, June). Tracking differential activation of primary and supplementary motor cortex across timing tasks: An fNIRS validation study. *20th Annual UC Systemwide Bioengineering Symposium*, Merced, CA.
20. **Rahimpour, A.**, Comstock, D., Pollonini, L., Balasubramaniam, R., & Bortfeld, H. (2019, May). Tracking differential activation of primary and supplementary motor cortex across timing tasks: An fNIRS validation study. *Berkeley/Stanford Developmental Psychology Symposium*, Berkeley, CA.
21. **Rahimpour, A.**, Pollonini, L., Comstock, D., Balasubramaniam, R., & Bortfeld, H. (2019, March). Tracking differential activation of primary and supplementary motor cortex across timing tasks: An fNIRS validation study. *26th Annual Meeting of the Cognitive Neuroscience Society*, San Francisco, CA.
22. Borjkhani, M., Khazenifard, A., & **Rahimpour, A.** (2014, May). Probabilistic study of different synchronization measures: Application to electroencephalographic signals. *Brain and Clinical Neuroscience Congress (BCNC)*, Tehran, Iran.

RESEARCH EXPERIENCE

Research Scientist and Postdoctoral Scholar, C-Brain Lab

2022 – present

Department of Psychiatry and Behavioral Sciences, Stanford University School of Medicine

- *Optical neuroimaging foundation model*: aggregating large-scale wearable fNIRS data and developing self-supervised (masked and contrastive) learning models for time series to enable scalable brain-state discovery across tasks, populations, and contexts.
- *Naturalistic at-home wearable fNIRS*: pilot of self-initiated, ambulatory prefrontal recordings during everyday activities paired with ecological momentary assessment (98 sessions over 24 days); task- and mood-dependent prefrontal functional connectivity.
- *fNIRS language mapping*: pilot comparing a whole-head NIRx system with a wearable prefrontal headband on a four-task covert language battery, toward a bedside alternative to Wada testing and fMRI for presurgical language lateralization.
- *ADHD deep phenotyping*: characterizing neurobiological variation in ADHD with multimodal wearable neuroimaging in structured and naturalistic settings.
- *Neurodiversity and decision-making under risk*: combining wearable neuroimaging with behavioral economics to study financial decision-making in neurodiverse populations.
- *Clinical neuroimaging in vulnerable populations*: identifying neurophysiological markers in PTSD, adverse childhood experiences, and Prader-Willi syndrome to inform intervention development.
- *Wearable fNIRS platform for precision mental health*: led beta-testing, quality control, and data processing for a custom wireless fNIRS headband with cloud-based analytics, designed for clinical monitoring and dense-sampling precision neuroimaging.
- *Neurofeedback for ADHD*: contributed to study design and fNIRS/fMRI data preprocessing for a personalized neurofeedback and neuromonitoring intervention to improve working memory in children with ADHD.

Graduate Student Researcher

2017 – 2022

Department of Psychological Sciences, University of California, Merced (Advisor: Heather Bortfeld)

- Designed and conducted EEG and fNIRS studies of rhythm perception, action-based timing, and auditory–motor network interaction.

R&D Biomedical Engineer

2016 – 2017

NR Sign Inc., Tehran, Iran

- Tested fNIRS hardware and processed its data; researched pain-transmission pathways; improved neuromuscular electrical stimulation protocols; designed experiments for EOG systems.

Collaborative Research Projects

Wearable fNIRS in an Alzheimer's disease cohort

2026 – present

Stanford Alzheimer's Disease Research Center (ADRC) — lead analyst

- Amyloid-characterized cohort. Feasibility and analysis of wearable fNIRS (37 channels, 108 recordings) during NIH Toolbox cognitive tasks, linked to amyloid and tau PET, plasma p-tau217, GFAP and NfL, and cognition; region-level activation and connectivity with biomarker and cognition associations.

SYSTEMAATICS: Childhood adversity and acute stress

2026 – present

Collaborative study — computational analysis lead and lead of the brain measurement team

- fNIRS during the Montreal Imaging Stress Task linking right lateral prefrontal activation to childhood-adversity load, task performance, and heart-rate variability (mixed-effects models, FDR correction, cross-validated prediction).

Vibrotactile gamma entrainment in amnesic MCI

2024

Stanford Bio-X Seed Grant proposal (Round 12), with P. Miri, M. Snyder, and S. M. H. Hosseini — lead for EEG and MRI outcome measures

- Double-blind randomized controlled trial of home-based vibrotactile stimulation for 40 Hz gamma entrainment; designed pre/post EEG entrainment and neuroimaging outcomes.

fNIRS replication of non-adjacent dependency learning

2026 – present

Replication of Lu & Mintz (2023), Exp. 1 — computational analysis collaboration

- Rebuilt the analysis pipeline (AR-IRLS first-level models, a priori ROI tests, FIR deconvolution) and ran a multiverse (specification-curve) analysis over 216 first-level pipelines and 8 channel-QC rules with permutation inference, label-blind pipeline selection, Bayes factors, and equivalence tests.

Fraction processing in South African schoolchildren

2023 – present

University of Johannesburg, South Africa — fNIRS data processing

Executive function in pediatric obstructive sleep apnea

2021 – 2022

University of Maryland School of Medicine, Baltimore, MD — pre- vs. post-tonsillectomy; fNIRS data preprocessing

Auditory stream segregation in cochlear implant users

2021 – 2022

Department of Otolaryngology, Keck School of Medicine, USC — experimental setup, data acquisition, and analysis

Software Development

Contributor, NIRS Toolbox (MATLAB)

2017 – present

Beta testing, troubleshooting, and development of an open-source toolbox for fNIRS analysis

Lead Developer, Precision Wearable (MATLAB)

in development

Open toolbox (110 functions, ~24,000 lines) for dense-sampling analysis of wearable fNIRS and EEG

- End-to-end pipeline with a graphical interface: 9 input formats, 15 quality metrics, 10 motion-correction methods, and AR-IRLS first-level models.
- Intensive-longitudinal methods: generalizability-theory reliability of within-person change vs. between-person differences, intraindividual variability, change detection, brain-EMA/wearable coupling, and growth-curve modeling.

Exploratory Neurotechnology Projects

Real-time EEG brain-computer interface for adaptive user experience

- Designed a closed-loop framework that estimates workload, attention, and engagement from bilateral ear-EEG to adapt presented content, supporting supervised, unsupervised, and reinforcement-learning paradigms (unpublished).

TEACHING EXPERIENCE

Workshops

2026 Instructor, workshop on MATLAB programming and analysis of longitudinal, dense-sampled neuroimaging time-series data, Division of Interdisciplinary Brain Sciences, Stanford Medicine

Guest Lecturer — Stanford University

2023 – 2026 PSYC 250: Neuroimaging Research Methods (Instructors: H. Hosseini, M. Saggar) — introduction to fNIRS and its analysis; annually, winter quarter

2023 – 2026 Summer SERI Talk Series — introduction to fNIRS and its clinical applications; annually, summer

Summer 2025 TechX Summer Camp — functional near-infrared spectroscopy and wearable neuroimaging (with C-Brain Lab team)

Summer 2025 Summer SERI Talk, Department of Linguistics — clinical applications of optical neuroimaging

Guest Lecturer — University of California, Merced

Summer 2019 Introduction to Neuropsychology (Instructor: A. Castillo)

Summer 2019 Introduction to Neuroimaging Methods (Instructor: A. Castillo)

Teaching Fellow — University of California, Merced

Fall 2021 Industrial/Organizational Psychology (Instructor: R. Avilla)

Spring 2021 Developmental Cognitive Neuroscience (Instructor: H. Bortfeld)

Teaching Assistant — University of California, Merced

Spring 2022 Cognitive Psychology (Instructor: R. Yancey)

Fall 2020 Physiological Psychology (Instructor: M. Altman)

Fall 2020 Industrial/Organizational Psychology (Instructor: R. Avilla)

Spring 2020 Cognitive Psychology (Instructor: R. Yancey)

Fall 2019 Cognitive Development (Instructor: R. Yancey)

Spring 2019 Developmental Cognitive Neuroscience (Instructor: H. Bortfeld)

Fall 2018 Alcohol, Drugs, and Behavior (Instructor: R. Yancey)

Spring 2018 Clinical Neuropsychology (Instructor: A. Khislavsky)

Fall 2017 Alcohol, Drugs, and Behavior (Instructor: R. Yancey)

Teaching Assistant — University of Tehran

Fall 2015 Statistical Pattern Recognition (Instructor: B. Nadjar Araabi)

Fall 2015 Introduction to Biomedical Instrumentation (Instructor: S. K. Setarehdan)

Teaching Assistant — Sahand University of Technology

Spring 2012 Design of Linear Control Systems (Instructor: A. Akbari)

Spring 2011 Advanced Electronics (Instructor: E. Najafi Aghdam)

Fall 2010 Engineering Mathematics (Instructor: Sadeghi Yazdankhah)

RESEARCH MENTORING

Mentored trainees and early-career researchers in fNIRS study design, experimental setup, data acquisition, and analysis (2019 – present); served as second reader for an undergraduate (B.S.) thesis. Current positions shown where known.

Trainees — Stanford University School of Medicine

- Arezoo Ghasemzadeh — Summer Intern (2026); undergraduate, Computational Cognitive Neuroscience, UC Davis
- Diana Anvari — Summer Intern (2026); high school student, The Nueva School
- Jennifer O’Grady Gonzalez De Ampuero — Summer Intern (2026); undergraduate, Human Biology
- Evelyn Williams — Summer Intern (2026); undergraduate, Santa Clara University
- Ivy Kilar — Summer Intern (2025); undergraduate, Neuroscience, Duke University
- Carmen Izuzkiza — Undergraduate Researcher; Human Biology, Stanford University
- Subhang Bhati — Research Assistant; Stanford Online High School
- Kenneth Nathanael Sarip — Undergraduate Researcher; EECS, UC Berkeley
- Laura Moreno Carbonell — Ph.D. student, Bioengineering, Stanford University
- Ester Patrizze Lopez Aguilar — Research Coordinator, Stanford School of Medicine
- Saba Faghihzadeh — Assistant Clinical Research Coordinator, Stanford School of Medicine
- Anna Aitken — B.S. Human Biology, Stanford; now M.P.H. student, Simon Fraser University
- Tulla Bee Picardi — B.S. Human Biology, Stanford; now M.D. student, University of South Florida

High School Students — CMP Academy

- Research Mentor, CMP Academy (2024 – present): one-on-one online research mentoring of high school students in neuroscience, cognitive science, and neurobiology; mentees have since been admitted to universities including Emory University, UCLA, USC, and Yale University.

Mentored student publications (mentor: A. Rahimpour Jounghani):

- Georgieva, V. (2025, September). The use of portable functional near-infrared spectroscopy (fNIRS) to study brain function in real-world sports environments. *Exploratio Journal*.
- Volvovsky, E. (2025, August). Portable fNIRS in sports: Measuring brain function in real-world settings. *Exploratio Journal*.

Early-Career Researchers (fNIRS methods training)

- Felix Hao Wang — then Assistant Professor, University of Nevada, Las Vegas; now Professor, Nanjing Normal University
- Samantha R. O’Connell — then Postdoctoral Researcher, Keck School of Medicine, USC; now Assistant Professor of Psychology, Wesleyan University
- Parisa Mobasheri — Ph.D. student, University of California, Merced
- Julianne Papadopoulos — then Master’s student, Keck School of Medicine, USC

PROFESSIONAL SERVICE

Editorial Roles

2024 – present Editorial Board Member, *Heliyon* (Cell Press)

2023 Guest Editor, Special Issue, *Frontiers in Human Neuroscience*

2023 Guest Editor, Special Issue, *Behavioural Brain Research*
 2021 Associate Editor, *Frontiers in Human Neuroscience*

Ad Hoc Journal Reviewer

Artificial Intelligence Review; Biomedical Engineering: Applications, Basis and Communications; Cerebral Cortex; Computer Methods and Programs in Biomedicine; Cortex; Frontiers in Human Neuroscience; Hearing Research; Heliyon; IEEE Access; IEEE Transactions on Neural Systems and Rehabilitation Engineering; Imaging Neuroscience; Infant Behavior and Development; Journal of Cognitive Neuroscience; Journal of Neural Engineering; Journal of Neuroimaging; Medical & Biological Engineering & Computing; Medical Hypotheses; NeuroImage; Proceedings of the National Academy of Sciences (PNAS); Scientific Reports; Sensors

Grant and Abstract Review

2023 Abstract Reviewer, Stanford Maternal & Child Health Research Institute (MCHRI) Symposium
 2023 Reviewer, Association for Psychological Science (APS) Annual Convention
 2021 Abstract Reviewer, Society for Functional Near-Infrared Spectroscopy (SfNIRS) Conference
 2019 Grant Reviewer, Research Project Grant, Leverhulme Trust (UK)

Conference Organization and Institutional Service

2025 – present Organizer and Lead, Weekly Journal Club, Division of Interdisciplinary Brain Sciences (DIBS), Stanford Medicine
 2021 – 2022 Organizing Committee Member and Session Chair, “fNIRS in Neurodevelopment Research” and “fNIRS Standards” sessions, SfNIRS 2021 Annual Meeting
 2020 Graduate Student Organizer, Developmental Psychology Colloquium Series, UC Merced
 2020 Outreach: fNIRS laboratory tours for high school students, Bortfeld Lab, UC Merced

TRANSLATIONAL AND ENTREPRENEURSHIP ACTIVITIES

Startup Pitches — Walnut Inc.

2026 Venture Summit West 2026 (youngStartup Ventures) — Top Innovator Honoree pitch
 2026 Ferment 2026, Stanford Biotech Group — selected startup pitch
 2025 Stanford Founders Demo Day — finalist pitch

Training and Programs

2026 StartX Accelerator (Stanford-affiliated) — Walnut Inc.; StartX founder and alumnus
 2025 Stanford Graduate School of Business Startup Garage (completed)
 2025 Stanford Venture Studio — featured project on neurofeedback
 2020 NSF Innovation Corps (I-Corps), UC Berkeley — selected proposal; advanced workshop participant

MEDIA COVERAGE AND SCIENCE COMMUNICATION

2025	<i>Stanford Daily</i> — coverage of Stanford Founders Demo Day (May)
2025	<i>Stanford Report</i> — neurofeedback and brain research in ADHD (March)
2025	<i>Stanford Science Highlights</i> — featured research article (March)
2026	Rahimpour Joungani, A. , Hosseini, H., Aitken, A., & Crawford, S. What happens to the brain after 15 minutes of scrolling? <i>Behind the Paper, Springer Nature Research Communities</i> (August 15)
2025	Rahimpour Joungani, A. , Hosseini, H., & Crawford, S. Mapping the mind at home: A wearable fNIRS platform for the future of precision mental health. <i>Behind the Paper, Springer Nature Research Communities</i> (June 13)
2024	Rahimpour Joungani, A. Revolutionizing ADHD treatment through neuromonitoring-guided working memory interventions. <i>Tianqiao and Chrissy Chen Institute</i> blog, featured by Stanford Chemical Engineering (November)

TECHNICAL SKILLS

Neuroimaging	fNIRS (wearable and lab-based systems; design, acquisition, preprocessing, GLM and functional connectivity analysis), EEG, fMRI, EOG; neurofeedback
Programming	Python, R, MATLAB (incl. NIRS Toolbox)
Statistical software	SAS, Stata, Mplus, SPSS, OpenBUGS
Statistics	General linear and logistic models, multilevel/mixed-effects models, structural equation modeling, factor analysis, longitudinal analysis, Bayesian modeling
Machine learning	Supervised (SVM, KNN, MLP, RBF networks); unsupervised (k-means, fuzzy c-means, self-organizing maps); dimensionality reduction (PCA, ICA); deep learning (CNN, RNN); bio-inspired optimization (genetic algorithms, ant colony and artificial bee colony optimization, cellular automata)

PROFESSIONAL MEMBERSHIPS

2019 – 2021	Cognitive Neuroscience Society (CNS)
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LANGUAGES

Persian (native) · English (fluent) · Arabic (elementary) · German (basic) · Turkish (basic)

REFERENCES

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